



## RS004 Week 16 — English Worksheet

**Topic:** Title · Level Select · Sound · Showcase (Final) · **Course:** Platformer Game · **Time:** about 45 minutes

This is the final week. You polish 15 weeks of work into a finished game: a **title screen**, a **level select** menu, and **sound** for jumps and coins. The whole thing runs on a **game state machine** that moves between title → level select → playing → game over / victory. Then you **present** it.

Keep these words handy: **state machine**, **title screen**, **level select**, **sound**, **showcase**.

### 1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
current_state = "title"
# "title" → "level_select" → "playing" → "gameover" / "victory" → "title"
```

The game starts in **"title"**. List one path it can take through the states to a win.

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```
if current_state == "title":
    if event.key == pygame.K_SPACE:
        current_state = "playing"
    elif event.key == pygame.K_l:
        current_state = "level_select"
```

On the title screen the player presses L. What screen comes next?

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```
if event.key == pygame.K_SPACE and on_ground:  
    velocity_y = JUMP_STRENGTH  
    jump_sound.play()
```

**When does the jump sound play? Could it play in the air? Why or why not?**

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## 2 · Map the States

Draw arrows (write them out) showing which state leads to which. Fill in the blanks.

```
title  --space--> _____  
title  --L-----> _____  
level_select  --Enter--> _____  
playing  --hit enemy--> _____  
playing  --clear last level--> _____  
gameover  --R--> _____
```

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### 3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

**Bug A** — Pressing space on the title should start the game. Right now space does nothing because the key is only checked while playing.

```
if current_state == "playing":  
    if event.key == pygame.K_SPACE:  
        velocity_y = JUMP_STRENGTH
```

**Hint:** the title screen also needs to react to space. Handle each state's keys.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — In the level select menu, ↑/↓ should move the highlight and wrap around the ends. Right now going past the last item crashes with an index error.

```
if event.key == pygame.K_DOWN:  
    selected_level = selected_level + 1
```

**Hint:** use `% len(LEVELS)` so the choice wraps back to the start.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — A coin pickup should play a sound, but the game is silent even though `coin_sound` exists.

```
for coin in coins[:]:
    if player.colliderect(coin):
        coins.remove(coin)
        score += 1
```

**Hint:** the sound is created but never played on pickup.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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## 4 · Polish Pass

Start from your working state machine. Add polish one piece at a time.

1. Add your own title text and a "press space to start" line. What does your title say?

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2. Add a jump sound *and* a coin sound. When does each play?

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3. **Stretch:** add a creator screen with your name, or a death sound, or a screen shake on Game Over.  
What did you add?

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## 5 · Finish & Showcase 📷

Bring it all together into your finished game: title → level select → at least three of your own levels → victory, with sound.

Send a **photo or video** of a full play-through (title to victory), then write your **showcase notes** below. Write 4 to 6 sentences.

The part of my game I am most proud of is ...

My favourite function or class is \_\_\_, and it works by ...

My game state machine moves from ... to ... to ...

The hardest bug I solved this term was ...

If I kept building, I would add ...

One thing I can do now that I couldn't 16 weeks ago is ...

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## 6 · Showcase Walkthrough 🎥

Record your final showcase video — this is the one you present. Teach your game to someone who has never seen it. Try to use these words: **state machine**, **title**, **level select**, **sound**, **function**.

1. Show the title screen, then level select.
2. Play through a full level with sound.
3. Show one function or class you are proud of and explain it.
4. Reach the victory screen.
5. Say the hardest bug you fixed and how.

**Write what you will say in your showcase.** Plan it here first.

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**Submit** ✓

Send this worksheet + your finished game video + your showcase notes to teacher on KakaoTalk.